14.1

1. 给定函数 求定义域，值域，判断定义域的开、闭性

23. $f(x, y) = \dfrac{1}{\sqrt{16 - x^2 - y^2}}$ 24.

14.2

1. 求给定函数极限

3. $\lim\limits_{(x, y)\to(3,4)} \sqrt{x^2 + y^2 - 1}$ 4.

Find the limits in Exercises 13–24 by

13. $\lim\limits_{\substack{(x, y)\to(1,1) \\ x\neq y}} \dfrac{x^2 - 2xy + y^2}{x - y}$ 14.

Find $\lim_{(x,y)\to(\infty,\infty)}(x^2 + y^2)\sin\frac{4}{x^2+y^2}$

Find $\lim\limits_{(x,y)\to(0,0)} \dfrac{\sin(xy)}{x}$

2. 求分段函数的极限

52. Let $f(x, y) = \begin{cases} x^2, & x \geq 0 \\ x^3, & x < 0 \end{cases}$.

Find the following limits.

a. $\lim_{(x, y)\to(3, -2)} f(x, y)$

b. $\lim_{(x, y)\to(-2, 1)} f(x, y)$

c. $\lim_{(x, y)\to(0,0)} f(x, y)$

14.3

1. 求给定函数的偏导数

19. $f(x, y) = x^y$

- 1. $z = e^{xy}\cos x$
- Answer $dz = e^{xy}(y\cos x - \sin x)dx + xe^{xy}\cos x dy$
- 2. $z = \sec(xy)$
- Answer $dz = \sec(xy)\cdot\tan(xy)\cdot(ydx + xdy)$
- 3. $u = x^{yz}, x > 0$
- Answer
- $du = yzx^{yz-1}dx + zx^{yz}\ln x dy + yx^{yz}\ln x dz$

2. 求分段函数的偏导数

60. $f(x, y) = \begin{cases} \dfrac{\sin(x^3 + y^4)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0), \end{cases}$

$\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ at $(0, 0)$

14.4

1. 利用链式法则求导数/偏导数

14. $\dfrac{dz}{dt}$ for $z = f(u, v, w), \quad u = g(t), \quad v = h(t), \quad w = k(t)$

15. $\dfrac{\partial w}{\partial}$ and $\dfrac{\partial w}{\partial}$ for $w = h(x, y, z), \quad x = f(u, v), \quad y = g(u, v)$

- Find $\dfrac{du}{dx}$ if $u = \dfrac{y-z}{1+a^2}e^{ax}$, $y = a\sin x$, $z = \cos x$.

4. $w = \ln(x^2 + y^2 + z^2), \quad x = \cos t, \quad y = \sin t, \quad z = 4\sqrt{t}$; $t = 3$

2. 隐函数求一阶/二阶偏导数

31. $\sin(x + y) + \sin(y + z) + \sin(x + z) = 0, \quad (\pi, \pi, \pi)$

32. $xe^y + ye^z + 2\ln x - 2 - 3\ln 2 = 0, \quad (1, \ln 2, \ln 3)$

Finding Partial Derivatives at Specified Points

- Class Work

- 1. Find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$ if

$$x^2 \sin y + e^x \arctan z - \sqrt{y}\ln z = 3.$$

- Answer: $2x\sin y + e^x \arctan z + \dfrac{e^x}{1+z^2}\dfrac{\partial z}{\partial x} - \dfrac{\sqrt{y}}{z}\dfrac{\partial z}{\partial x} = 0$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{z(1+z^2)(2x\sin y + e^x \arctan z)}{(1+z^2)\sqrt{y} - ze^x},$$

$$x^2\cos y + \frac{e^x}{1+z^2}\frac{\partial z}{\partial y} - \frac{\ln z}{2\sqrt{y}} - \frac{\sqrt{y}}{z}\frac{\partial z}{\partial x} = 0 \Rightarrow \frac{\partial z}{\partial y} = \frac{z(1+z^2)(2x^2\sqrt{y}\cos y - \ln z)}{2(1+z^2)y - 2z\sqrt{y}e^x}.$$

- 2. Find $\dfrac{\partial^2 z}{\partial x \partial y}$ if $e^z - xy + z - 1 = 0$.

- Answer: $\dfrac{\partial z}{\partial y} = \dfrac{x}{e^z+1}$, $\dfrac{\partial^2 z}{\partial x \partial y} = \dfrac{\partial}{\partial y}\left(\dfrac{\partial z}{\partial x}\right) = \dfrac{\partial}{\partial y}\left(\dfrac{y}{e^z+1}\right) = \dfrac{(e^z+1) - ye^z\dfrac{\partial z}{\partial y}}{(e^z+1)^2} = \dfrac{(e^z+1)^2 - xye^z}{(e^z+1)^3}$

23. $f(x, y) = \dfrac{1}{\sqrt{16 - x^2 - y^2}}$

§14.5

1. 求方向导数

EXAMPLE 1 Using the definition, find the derivative of

$$f(x, y) = x^2 + xy$$

at $P_0(1, 2)$ in the direction of the unit vector $\mathbf{u} = \left(1/\sqrt{2}\right)\mathbf{i} + \left(1/\sqrt{2}\right)\mathbf{j}$.

P11

Class Work

P14

1. Find the directional derivative of $f(x, y, z) = x^3 - xy^2 - z$ at the point (1,1,0) in the direction of vector $\boldsymbol{v} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$.
- Answer 4/7

2. Find the directional derivative of $f(x, y, z) = \ln\left(x + \sqrt{y^2 + z^2}\right)$ at the point (1,0,1) in the direction of vector $\boldsymbol{v} = 3\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$.
- Answer 1/2

Exercises 14.5

Calculating Gradients

In Exercises 1–6, find the gradient of the function at the given point. Then sketch the gradient together with the level curve that passes through the point.

1. $f(x, y) = y - x$, $(2, 1)$
2. $f(x, y) = \ln(x^2 + y^2)$, $(1, 1)$
3. $g(x, y) = xy^2$, $(2, -1)$
4. $g(x, y) = \frac{x^2}{2} - \frac{y^2}{2}$, $(\sqrt{2}, 1)$
5. $f(x, y) = \sqrt{2x + 3y}$, $(-1, 2)$
6. $f(x, y) = \tan^{-1}\frac{\sqrt{x}}{y}$, $(4, -2)$

In Exercises 7–10, find ∇f at the given point.

7. $f(x, y, z) = x^2 + y^2 - 2z^2 + z \ln x$, $(1, 1, 1)$
8. $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z + \tan^{-1} xz$, $(1, 1, 1)$
9. $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2} + \ln(xyz)$, $(-1, 2, -2)$
10. $f(x, y, z) = e^{x+y}\cos z + (y + 1)\sin^{-1} x$, $(0, 0, \pi/6)$

Finding Directional Derivatives

In Exercises 11–18, find the derivative of the function at P_0 in the direction of $\mathbf{u}$.

11. $f(x, y) = 2xy - 3y^2$, $P_0(5, 5)$, $\mathbf{u} = 4\mathbf{i} + 3\mathbf{j}$
12. $f(x, y) = 2x^2 + y^2$, $P_0(-1, 1)$, $\mathbf{u} = 3\mathbf{i} - 4\mathbf{j}$
13. $g(x, y) = \frac{x - y}{xy + 2}$, $P_0(1, -1)$, $\mathbf{u} = 12\mathbf{i} + 5\mathbf{j}$
14. $h(x, y) = \tan^{-1}(y/x) + \sqrt{3}\sin^{-1}(xy/2)$, $P_0(1, 1)$, $\mathbf{u} = 3\mathbf{i} - 2\mathbf{i}$

P17

EXAMPLE 2 Find the derivative of $f(x, y) = xe^y + \cos(xy)$ at the point $(2, 0)$ in the direction of $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$.

2. 求切平面及法线 (§14.6)

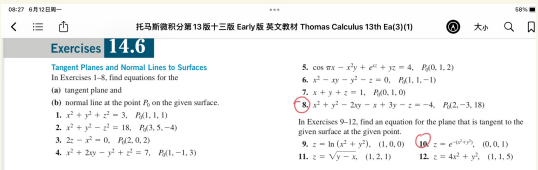
Example 1 Find the tangent plane and normal line of the surface $F(x, y, z) = x^2 + y^2 + Z - 9 = 0$ at the point $P_0(1, 2, 4)$.

P35

Example 2 Find the tangent plane to the surface $z = x\cos y - ye^x$ at (0, 0, 0).

P37

Hint $F(x,y,z) = x\cos y - ye^x - z$.



3、求参数方程

Example 3 The surfaces

$$f(x, y, z) = x^2 + y^2 - 2 = 0$$

and

$$g(x, y, z) = x + z - 4 = 0$$

meet in an ellipse E. Find parametric equations for the line tangent to E at the point $P_0(1, 1, 3)$.

4、偏导的几何意义

Example 5 Find the linearization of

$$f(x, y) = x^2 - xy + \frac{1}{2}y^2 + 3$$

at the point (3, 2).

14.7

1、利用定义求极值

EXAMPLE 1 Find the local extreme values of $f(x, y) = x^2 + y^2 - 4y + 9$.

Example 2 Find the local extreme values (if any) of $f(x, y) = y^2 - x^2$.

2、二阶判别法求极值

Example 4

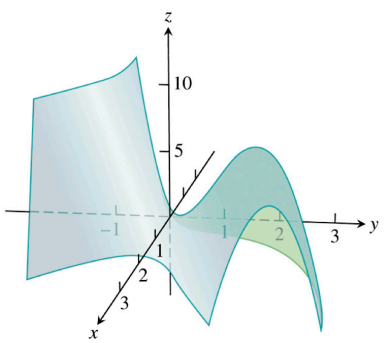


FIGURE 14.45 The surface $z = 3y^2 - 2y^3 - 3x^2 + 6xy$ has a saddle point at the origin and a local maximum at the point (2, 2) (Example 4).

Finding Local Extrema

Find all the local maxima, local minima, and saddle points of the functions in Exercises 1–30.

1. $f(x, y) = x^2 + xy + y^2 + 3x - 3y + 4$
2. $f(x, y) = 2xy - 5x^2 - 2y^2 + 4x + 4y - 4$
3. $f(x, y) = x^2 + xy + 3x + 2y + 5$
4. $f(x, y) = 5xy - 7x^2 + 3x - 6y + 2$
5. $f(x, y) = 2xy - x^2 - 2y^2 + 3x + 4$
6. $f(x, y) = x^2 - 4xy + y^2 + 6y + 2$
7. $f(x, y) = 2x^2 + 3xy + 4y^2 - 5x + 2y$
8. $f(x, y) = x^2 - 2xy + 2y^2 - 2x + 2y + 1$
9. $f(x, y) = x^2 - y^2 - 2x + 4y + 6$
10. $f(x, y) = x^2 + 2xy$
11. $f(x, y) = \sqrt{56x^2 - 8y^2 - 16x - 31} + 1 - 8x$
12. $f(x, y) = 1 - \sqrt[3]{x^2 + y^2}$
13. $f(x, y) = x^3 - y^3 - 2xy + 6$
14. $f(x, y) = x^3 + 3xy + y^3$
15. $f(x, y) = 6x^2 - 2x^3 + 3y^2 + 6xy$
16. $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2 - 8$
17. $f(x, y) = x^3 + 3xy^2 - 15x + y^3 - 15y$
18. $f(x, y) = 2x^3 + 2y^3 - 9x^2 + 3y^2 - 12y$
19. $f(x, y) = 4xy - x^4 - y^4$
20. $f(x, y) = x^4 + y^4 + 4xy$
21. $f(x, y) = \dfrac{1}{x^2 + y^2 - 1}$
22. $f(x, y) = \dfrac{1}{x} + xy + \dfrac{1}{y}$

Class Work

Find the extreme, if any, of function f defined by $f(x, y) = x^2 + xy + y^2 + x - y + 1$.

Example 6

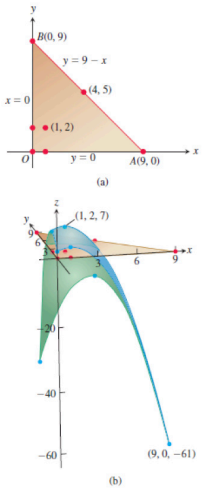


FIGURE 14.50 (a) This triangular region is the domain of the function in Example 6. (b) The graph of the function in Example 6. The blue points are the candidates for maxima or minima.

Finding Absolute Extrema

In Exercises 31–38, find the absolute maxima and minima of the functions on the given domains.

31. $f(x, y) = 2x^2 - 4x + y^2 - 4y + 1$ on the closed triangular plate bounded by the lines $x = 0$, $y = 2$, $y = 2x$ in the first quadrant
32. $D(x, y) = x^2 - xy + y^2 + 1$ on the closed triangular plate in the first quadrant bounded by the lines $x = 0$, $y = 4$, $y = x$
33. $f(x, y) = x^2 + y^2$ on the closed triangular plate bounded by the lines $x = 0$, $y = 0$, $y + 2x = 2$ in the first quadrant
34. $T(x, y) = x^2 + xy + y^2 - 6x$ on the rectangular plate $0 \le x \le 5$, $-3 \le y \le 3$
35. $T(x, y) = x^2 + xy + y^2 - 6x + 2$ on the rectangular plate $0 \le x \le 5$, $-3 \le y \le 0$
36. $f(x, y) = 48xy - 32x^3 - 24y^2$ on the rectangular plate $0 \le x \le 1$, $0 \le y \le 1$
37. $f(x, y) = (4x - x^2)\cos y$ on the rectangular plate $1 \le x \le 3$, $-\pi/4 \le y \le \pi/4$ (see accompanying figure)

14.8

1、约束转换为无约束求极值

Example 1 Find the point $P(x, y, z)$ on the plane $2x + y - z - 5 = 0$ that is closest to the origin.

Example 2 Find the point $P(x, y, z)$ on the hyperbolic cylinder $x^2 - z^2 - 1 = 0$ that is closest to the origin.

2、拉格朗日法求极值

Example 3 Find the greatest and smallest values that the function $f(x, y) = xy$ takes on the ellipse

$$\frac{x^2}{8} + \frac{y^2}{2} = 1.$$

Example 4 Find the maximum and minimum values that the function $f(x, y) = 3x + 4y$ takes on the circle $x^2 + y^2 = 1$.

Class Work

1. Find the maximum and minimum values of $f(x, y) = xy$ on the closed and bounded region D: $x^2 + y^2 \leq 1$.

1

2. Find the point on the paraboloid $z = x^2 + y^2$ that is closest to (0, 0, 2). What is the minimum distance?

8. **Extrema on a curve** Find the points on the curve $x^2 + xy + y^2 = 1$ in the xy-plane that are nearest to and farthest from the origin.

9. **Minimum surface area with fixed volume** Find the dimensions of the closed right circular cylindrical can of smallest surface area whose volume is 16π cm^3.

§14.8

Three Independent Variables with One Constraint

17. **Minimum distance to a point** Find the point on the plane $x + 2y + 3z = 13$ closest to the point $(1, 1, 1)$.

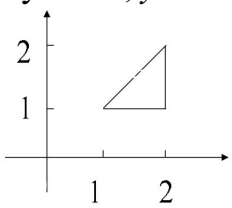
18. **Maximum distance to a point** Find the point on the sphere $x^2 + y^2 + z^2 = 4$ farthest from the point $(1, -1, 1)$.

19. **Minimum distance to the origin** Find the minimum distance

15.2

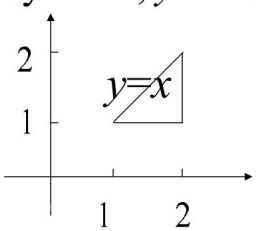
1、x型积分区域

Suppl. Example 1 Find $\iint_R xy\,dA$, R is bounded by $x = 2$, $y = 1$ and $y = x$.



2、y型区域

Suppl. Example 1 Find $\iint_R xy\,dA$, R is bounded by $x = 2$, $y = 1$ and $y = x$.

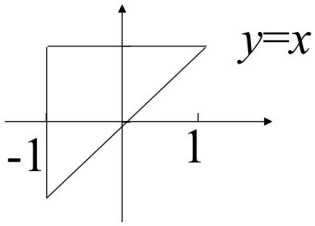


3、即是x型区域也是y型区域

EXAMPLE 1 Find the volume of the prism whose base is the triangle in the xy-plane bounded by the x-axis and the lines $y = x$ and $x = 1$ and whose top lies in the plane

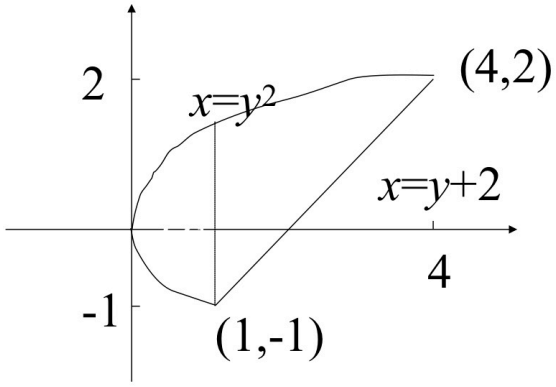
$$z = f(x, y) = 3 - x - y.$$

❖ **Supple. Example 2** $\iint_R x\sqrt{1-x^2+y^2}\,dA$



R is bounded by
$x = -1$, $y = 1$ and $y = x$.

❖ **Supple. Example 3** $\iint_R xy\,dA$

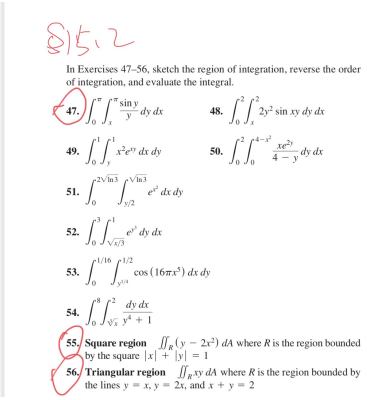


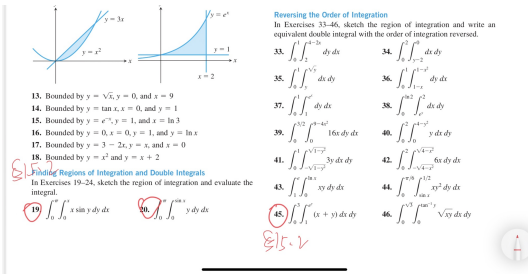
R is bounded by $y^2 = x$, and $y = x - 2$

3、积分顺序可交换

- **Supple. Example 4** Exchange the order of integration.

$$\int_0^2 \int_{y^2}^{2y} f(x,y)\mathrm{d}x\mathrm{d}y$$



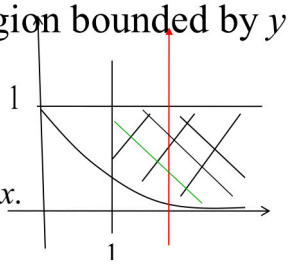


4、无界区域上的积分

4. Integrals over Unbounded Regions (Exercises 15.2)

- Used extensively in Probability and Statistics
- 69. $\int\int_R \frac{1}{x^3 y}\mathrm{d}A$
- R is an unbounded region bounded by $y=1$,
- $x=1$, $y=\mathrm{e}^{-x}$.

$$\int\int_R \frac{1}{x^3 y}\mathrm{d}A \overset{R:x-\text{type}}{=} \int_1^{\infty}\int_{\mathrm{e}^{-x}}^{1}\frac{1}{x^3 y}\mathrm{d}y\mathrm{d}x.$$



15.3

1、求积分区域面积

EXAMPLE 1 Find the area of the region R bounded by $y = x$ and $y = x^2$ in the first quadrant.

14. $\displaystyle\int_0^3 \int_{-x}^{x(2-x)} dy\, dx$

2、求积分函数在积分区域的平均值

EXAMPLE 3 Find the average value of $f(x, y) = x \cos xy$ over the rectangle R: $0 \le x \le \pi$, $0 \le y \le 1$.

15.4

1、利用极坐标变换求积分函数

- **Example 3(Sec.15.4)** Evaluate $\iint_R e^{x^2+y^2} dA$, where R is the semicircular region bounded by x-axis and the curve $y = \sqrt{1-x^2}$.

- Example 4

$$\int_0^1 \left[\int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy \right] dx$$

- **2**. Evaluate $\iint_R (x^2 + 2x^3 + \sin y^3 + 19)\, dx dy$, where R is the plane region $x^2 + y^2 \le 1$.

- **1**. Evaluate $\iint_R \arctan \frac{y}{x} dx dy$,where
- R is in the first quadrant bounded by x-axis, $x^2 + y^2 = 1$, $x^2 + y^2 = 4$, and the line $y = x$.

- Answer: $\frac{3\pi^2}{64}$

Evaluating Polar Integrals

In Exercises 9–22, change the Cartesian integral into an equivalent polar integral. Then evaluate the polar integral.

9. $\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}} dy\,dx$ **10.** $\int_{0}^{1}\int_{0}^{\sqrt{1-y^2}} (x^2+y^2)\,dx\,dy$

11. $\int_{0}^{2}\int_{0}^{\sqrt{4-y^2}} (x^2+y^2)\,dx\,dy$

12. $\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dy\,dx$ **13.** $\int_{0}^{6}\int_{0}^{y} x\,dx\,dy$

14. $\int_{0}^{2}\int_{0}^{x} y\,dy\,dx$ **15.** $\int_{1}^{\sqrt{3}}\int_{1}^{x} dy\,dx$

2、极坐标变换求积分与直角坐标求积分相互转换

In Exercises 23–26, sketch the region of integration and convert each polar integral or sum of integrals to a Cartesian integral or sum of integrals. Do not evaluate the integrals.

23. $\int_{0}^{\pi/2}\int_{0}^{1} r^3 \sin\theta\cos\theta\,dr\,d\theta$

24. $\int_{\pi/6}^{\pi/2}\int_{1}^{\csc\theta} r^2\cos\theta\,dr\,d\theta$

15、5

1、计算三重积分

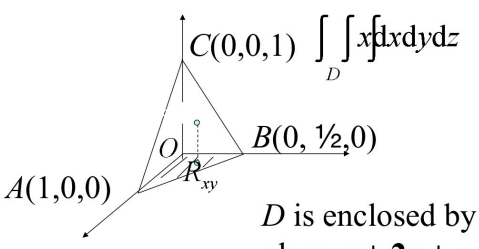
(Tongji P167, 5-8, 9(2)) **Home Work**

1. Evaluate the integral $\iiint_D \frac{1}{(1+x+y+z)^3}\,dxdydz$

where D is enclosed by the plane $x+y+z=1$ and three coordinates planes

Answer: $\frac{1}{2}\left(\ln 2-\frac{5}{8}\right)$

Suppl. Example 1 Evaluate the integral $\iiint_D x\,dxdydz$



D is enclosed by the plane $x+2y+z=1$ and three coordinates planes

3.Evaluate the integral

$$\iiint_D xz\,dV$$

where D is enclosed by the para ' bolic ' cyclinder $y = x^2$ and three planes $z = 0$, $z = y$, $y = 1$.

P93

Answer: 0

2.Evaluate the integral

$$\iiint_D xyz\,dV$$

where D is enclosed by the sphere $x^2 + y^2 + z^2 = 1$ and three coordinates planes.

P92

Answer: $\frac{1}{48}$

5. Evaluate the integral $\iiint_D (x + y + z)\,dV$

where D is enclosed by the plane $x + y + z = 1$ and three coordinates planes

P95

Answer: 1/8

4.Evaluate the integral

$$\iiint_D z\,dV$$

where D is enclosed by the cone $z = \frac{h}{R}\sqrt{x^2 + y^2}$ and the plane $z = h$ ($R > 0$, $h > 0$).

P94

Answer: $\frac{\pi R^2 h^2}{4}$

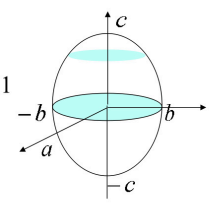
Suppl. Example 2 Evaluate $\iiint_D z^2\,dV$

P98

$$\overset{xy\text{-type?}}{=} \iint_{D_{xy}} \left[\int_{-c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}}^{+c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}} z^2\,dz \right] dx\,dy$$

$$= \frac{2c^3}{3} \iint_{D_{xy}} \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}\right)^{\frac{3}{2}} dx\,dy\ ?$$

$$D: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \le 1$$



$$R_{xy}: \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1 - \frac{z^2}{c^2}$$

6.Evaluate the integral

$$\iiint_D (x^2 + y^2)\,dV$$

where D is the solid enclosed by $x^2 + y^2 = 2z$ and the plane $z = 2$.

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Answer: $\frac{16\pi}{3}$

2、利用三重积分算体积

Home Work:

(2018)Let D be the closed region bounded by the surfaces $x^2 + y^2 = 1$, $z = 1$, and $z = 0$, then the integral

$$\iiint_D \left(3 + e^{z^3}\tan x^2 y^3\right) dV = (\quad)$$

(A) 0 (B) 3π (C) π (D) 3

Example 2 (**Class Work**)

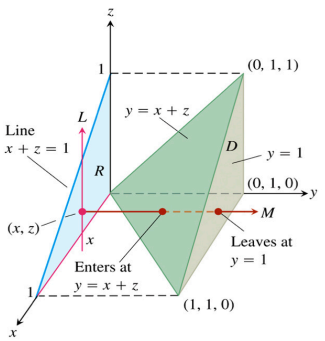


FIGURE 15.31 Finding the limits of integration for evaluating the triple integral of a function defined over the tetrahedron D (Examples 2 and 3).